

MINI-ENGAGEMENT PLAN – *Team CrossRiverPack*

Agricultural Extension RAG: Smart Retrieval for Farmers

1. Project Context

This engagement plan follows on from a stakeholder map built using Wakunuma and Jiya’s method [1] for the Agricultural Extension RAG project: an offline retrieval engine that returns the top five most relevant documents from a knowledge base of agricultural extension materials in response to smallholder farmers’ questions about ten common West African crops such as maize, cassava, cowpea, ginger, pepper, cocoa, groundnut, rice, tomato, and yam. The current system compares BM25, dense retrieval (sentence-transformer embeddings), a BM25–dense hybrid, and cross-encoder reranking, with a dense-retrieval configuration using intfloat/e5-base-v2 (title + title + text document representation) selected as the best-performing approach, reaching an nDCG@5 of 0.796 on the development set and 0.771 on the held-out validation set.

Moving from stakeholder analysis to action, this plan selects two stakeholder groups from that map and designs a concrete engagement method for each, following the Action Catalogue [3] and guided by the risks illustrated in the Nigerian chatbot case study, where excluding groups such as women’s rights advocates and service workers reinforced harmful stereotypes and missed real risks [2].

2. Select Stakeholder Groups

- Group 1: Women smallholder farmers across the target crop-growing regions of West Africa.
- Group 2: Agricultural extension officers who author, translate, and field-test the extension manuals in the document base.

3. Group 1: Women Smallholder Farmers

3.1. What Matters Most

Women smallholder farmers grow much of West Africa’s cassava, cowpea, groundnut, rice, and tomato, yet receive less formal extension contact, credit, and digital access than men, and often farm around household duties that limit when they can engage with new tools. What matters most to them is that this retrieval system speaks in language they can act on immediately: short, practical answers about pest and disease symptoms, planting windows, and soil care, available offline given patchy connectivity and shared phones. They also value privacy around admitting crop failure, and want the system to reflect their actual farming roles rather than male-headed household norms found in many extension manuals.

3.2. Potential Harms and Benefits

Potential harms

- Queries and outputs framed around male-headed farming norms.
- Loss of privacy when admitting crop losses or financial strain.
- Low-literacy or dialect phrasing misread by the retrieval model, surfacing irrelevant top-5 documents.
- Further marginalization if design sessions are scheduled at times that exclude caregivers.

Potential benefits

- Faster, private access to agronomic advice without waiting for a scarce extension visit.
- A query set and interface genuinely built around how women farmers actually ask questions.
- A documented channel for their priorities to shape which crops and topics the document base emphasizes.

3.3. Engagement Method: Participatory Design

Participatory Design is the strongest fit for engaging women smallholder farmers because it positions them as co-creators of the retrieval system rather than end-of-pipeline testers, which directly addresses the risk the problem statement and prior case study both raise: a system built without their input can misjudge relevance or reinforce exclusion. In practice, this means running a short series of workshops with existing women's farming cooperatives or village savings groups, held at times and locations that fit around caregiving and market days, and facilitated in the relevant local language rather than English or French, since language choice alone can determine whether a session is genuinely inclusive or merely symbolic participation.

Each session would ask participants to pose real questions about the ten target crops in their own words, giving the project authentic query phrasing to compare against the more formal, textbook-style language currently used in **train_queries** and **test_queries**, which was largely shaped by document authors rather than end users. Participants would then review the top-five documents the system actually returns for their own questions, flagging cases where results feel generic, outdated, or oriented toward commercial or male-headed farming contexts, and naming topics they wish the document base covered but does not, such as intercropping practices common on the smaller, more diverse plots women tend to manage.

Because Participatory Design is iterative rather than a single consultation, it also lets the project check whether earlier feedback was actually acted on, which builds trust that is otherwise easy to lose with a one-off session, especially with a group historically underrepresented in formal agricultural research and technology design. This matters given the Nigerian chatbot example, where excluding women's rights advocates and service workers from design contributed to outputs that reinforced harmful stereotypes; the same risk applies here if the training queries and relevance judgments implicitly assume a male commercial farmer as the default user.

A further strength of Participatory Design is that it produces concrete artifacts rather than just opinions: annotated query lists, marked-up document rankings, and notes on preferred phrasing, all of which map directly onto components already in the pipeline, namely the query sets and the **qrels** used to compute **nDCG@5**. Compared with a lighter-touch method such as a single survey, this hands-on format better suits an offline, lower-literacy context where a written feedback form would exclude the very participants the project most needs to reach. Compared with a heavier deliberative format built for contested public-policy disagreements, it is proportionate to a technical co-design task rather than a values debate.

Running two or three short workshops spaced across the project timeline, rather than one long session, respects participants' time while letting the design visibly evolve in response to what farmers say, which is

the core promise of Participatory Design as an engagement method and the clearest route from stakeholder mapping to actual influence over the system.

4. Group 2: Agricultural Extension Officers

4.1. What Matters Most

Agricultural extension officers' author, translate, and field-test the manuals this system draws on, so what matters most is that the tool strengthens rather than replaces their advisory role. They want confidence that retrieved documents are current, regionally accurate for the ten target crops, and correctly ranked, since a wrongly surfaced pest or disease recommendation could damage farmer trust in both the tool and the officer who endorsed it. They also care about workload: a system that quickly surfaces the right reference is valuable, but one needing constant correction becomes extra work. Recognition of their expertise in shaping the document set and evaluation criteria matters as much as the technology itself.

4.2. Potential Harms and Benefits

Potential harms

- Erosion of professional authority if the tool is deployed without officer input.
- Reputational risk if the system attaches an officer's credibility to a poorly ranked or outdated document.
- Extra burden if officers must repeatedly flag the same retrieval errors with no process for the fixes to stick.

Potential benefits

- A faster reference tool during field visits and training sessions.
- A structured channel to keep the underlying document base current as agronomic guidance changes.
- Recognition as co-owners of the evaluation criteria (the qrels) that determine what counts as a "relevant" document.

4.3. Engagement Method: Standing Expert Advisory Committee

A standing User or Expert Advisory Committee made up of agricultural extension officers is the better fit for this stakeholder group because their relationship to the project is technical and professional rather than experiential, and the risks they raise are ongoing rather than one-time. Extension officers are best positioned to judge whether a retrieved document is actually agronomically correct for a given crop, growth stage, and region, which is a form of domain expertise the project cannot substitute with farmer feedback alone. A committee format suits this because it creates a recurring touchpoint, for example quarterly reviews, rather than a single workshop, matching the fact that the document base, the relevance judgments, and the model itself will keep changing as the project moves from the current e5-base-v2 dense-retrieval baseline toward further refinement and eventual field deployment.

In each session, officers would audit a sample of query-to-document rankings, checking specifically for cases where a highly ranked document is topically related but agronomically wrong for the query's crop or

region, since these near-miss errors are the ones most likely to mislead a farmer acting on the advice without realizing it is subtly incorrect. They would also review gaps in the underlying document set, flagging crops or topics among the ten target crops that are thinly covered, and helping curate or contribute additional extension material where the corpus is weak, directly feeding the documents.csv file the retrieval system searches over. Involving officers in reviewing and correcting the **qrels_train** relevance labels is especially valuable, since these labels define what counts as “relevant” for every nDCG@5 score reported in the project, and officers are the only stakeholder qualified to make that judgment reliably rather than by proxy. An advisory committee also protects officers’ professional standing: rather than having a tool deployed over their heads that could contradict their advice in front of a farmer, they become named contributors to its accuracy, which the problem statement’s own goal of supporting rather than replacing extension work depends on in practice.

This method is a better fit than Participatory Design here because officers do not need help articulating their needs in accessible language; they need a structured, recurring role with real influence over technical decisions, which a standing committee is designed to provide over a distributed dialogue or one-off consultation. It is also more sustainable than a single expert review conducted at project launch, since agronomic best practice and document currency will keep shifting after deployment, and the retrieval system needs an ongoing mechanism to keep pace rather than freezing officer input at one point early in the project’s development and never revisiting it again.

5. Influence on Dataset and Model Design

Farmer-facing Participatory Design sessions would expand **train_queries** and **test_queries** with authentic local-language phrasing, correcting the current gap between formal agronomic wording and how farmers actually ask questions, which should improve the e5-base-v2 dense-retrieval model’s generalization beyond textbook-style queries. The extension-officer committee would directly refine **qrels_train** relevance labels and flag gaps or outdated entries in documents.csv, changing both what nDCG@5 measures and what the system has available to retrieve. Together, these methods shift the pipeline from technically accurate but narrowly validated toward a corpus and evaluation standard that reflects real farmer language and verified regional agronomic accuracy.

References

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